KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5282 | 5 Credits: 4 | Program Core Course | Source-grounded

Network Security

 To know the concepts & basic vocabulary of network security & security

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5282 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5282.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Cyber Forensics And Information Security
Course code	5282
Course title	Network Security
Semester	5 Credits: 4
Course category	Program Core Course
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

30 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To know the concepts & basic vocabulary of network security & security
- infrastructures.
- To identify the various cryptographic protocols along with hardware & software
- To understand WAP security & security issues with WTLS.
- To learn how Network Access Controls can be implemented.
- To identify the application and use of Firewalls.

Course outcomes

CO	Official outcome	Study level
CO1	protocols, reference models and fundamental 15 Understanding concepts of Network security. Understand the security concerns in Data link	See official syllabus
CO2	layer and the protocols for Wireless security and 14 Understanding IP security. Describe the functioning of most common	See official syllabus
CO3	security protocols employed at Transport Layer 15 Understanding and Application layer. Understand the importance of Network Access	See official syllabus
CO4	Control and the role of Firewalls in Network 14 Understanding security.	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3 2
CO3	CO3 3 2

Row	Extracted official mapping
CO4	CO4 3 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	and fundamental concepts of Network security.	7	As specified
Module 2	Wireless security and IP security.	7	As specified
Module 3	Transport Layer and Application layer.	9	As specified
Module 4	Firewalls in Network security.	7	As specified

MODULE 1

and fundamental concepts of Network security.

This module is presented from the official syllabus scope for Network Security. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: and fundamental concepts of Network security.
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

2 Understanding on the topology. Explain computer networks classification based

2 Understanding on the topology. Explain computer networks classification based is an official topic in Module 1 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding on the topology. Explain computer networks classification based using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding on the topology. explain computer networks classification based should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding on the topology. Explain computer networks classification based means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 2 Understanding on the topology. Explain computer networks classification based □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

2 Understanding on the scale and the distance they cover. Explain Network Reference model, ISO/OSI

2 Understanding on the scale and the distance they cover. Explain Network Reference model, ISO/OSI is an official topic in Module 1 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding on the scale and the distance they cover. Explain Network Reference model, ISO/OSI using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding on the scale and the distance they cover. explain network reference model, iso/osi should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding on the scale and the distance they cover. Explain Network Reference model, ISO/OSI means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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3 Understanding reference model and TCP/IP reference model.

3 Understanding reference model and TCP/IP reference model. is an official topic in Module 1 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding reference model and TCP/IP reference model. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding reference model and tcp/ip reference model. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding reference model and TCP/IP reference model. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-[illegible] 3 Understanding reference model and TCP/IP reference model. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain the OSI security architecture in detail. 2 Understanding Explain the needs for Network security and the

Explain the OSI security architecture in detail. 2 Understanding Explain the needs for Network security and the is an official topic in Module 1 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the OSI security architecture in detail. 2 Understanding Explain the needs for Network security and the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the osi security architecture in detail. 2 understanding explain the needs for network security and the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the OSI security architecture in detail. 2 Understanding Explain the needs for Network security and the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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2 fundamental concepts of Network Security. Understanding Explain the goals of Network security.

2 fundamental concepts of Network Security. Understanding Explain the goals of Network security. is an official topic in Module 1 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 fundamental concepts of Network Security. Understanding Explain the goals of Network security. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 fundamental concepts of network security. understanding explain the goals of network security. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 fundamental concepts of Network Security. Understanding Explain the goals of Network security. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Describe the different types of Network 2 Understanding security.

Describe the different types of Network 2 Understanding security. is an official topic in Module 1 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the different types of Network 2 Understanding security. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the different types of network 2 understanding security. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the different types of Network 2 Understanding security. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-[illegible] Describe the different types of Network 2 Understanding security. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain the basic model for Network Security. 2 Understanding Network Fundamentals : Computer Networks – Bus topology, Star topology, Ring topology , Tree topology, Mesh topology , Categories of Networks – LAN,MAN,WAN. Network Reference models – ISO/OSI reference model, TCP/IP reference model, The OSI security architecture. Network Security Overview: Fundamental concepts, Goals of Network Security, Types of Network security, a model for Network Security. Understand the security concerns in Data link layer and the protocols for

Explain the basic model for Network Security. 2 Understanding Network Fundamentals : Computer Networks – Bus topology, Star topology, Ring topology , Tree topology, Mesh topology , Categories of Networks – LAN,MAN,WAN. Network Reference models – ISO/OSI reference model, TCP/IP reference model, The OSI security architecture. Network Security Overview: Fundamental concepts, Goals of Network Security, Types of Network security, a model for Network Security. Understand the security concerns in Data link layer and the protocols for is an official topic in Module 1 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the basic model for Network Security. 2 Understanding Network Fundamentals : Computer Networks – Bus topology, Star topology, Ring topology , Tree topology, Mesh topology , Categories of Networks – LAN,MAN,WAN. Network Reference models – ISO/OSI reference model, TCP/IP reference model, The OSI security architecture. Network Security Overview: Fundamental concepts, Goals of Network Security, Types of Network security, a model for Network Security. Understand the security concerns in Data link layer and the protocols for using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the basic model for network security. 2 understanding network fundamentals : computer networks – bus topology, star topology, ring topology , tree topology, mesh topology , categories of networks – lan,man,wan. network reference models – iso/osi reference model, tcp/ip reference model, the osi security architecture. network security overview: fundamental concepts, goals of network security, types of network security, a model for network security. understand the security concerns in data link layer and the protocols for should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the basic model for Network Security. 2 Understanding Network Fundamentals : Computer Networks – Bus topology, Star topology, Ring topology , Tree topology, Mesh topology , Categories of Networks – LAN,MAN,WAN. Network Reference models – ISO/OSI reference model, TCP/IP reference model, The OSI security architecture. Network Security Overview: Fundamental concepts, Goals of Network Security, Types of Network security, a model for Network Security. Understand the security concerns in Data link layer and the protocols for means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Explain the basic model for Network Security. 2 Understanding Network Fundamentals : Computer Networks – Bus topology, Star topology, Ring topology , Tree topology, Mesh topology , Categories of Networks – LAN,MAN,WAN. Network Reference models – ISO/OSI reference model, TCP/IP reference model, The OSI security architecture. Network Security Overview: Fundamental concepts, Goals of Network Security, Types of Network security, a model for Network Security. Understand the security concerns in Data link layer and the protocols for □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

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and fundamental concepts of Network security.
              Explain computer networks classification based
   M1.01                                                        2
Understanding
              on the topology.
              Explain computer networks classification based
   M1.02                                                        2
Understanding
              on the scale and the distance they cover.
              Explain Network Reference model, ISO/OSI
   M1.03                                                        3
Understanding
              reference model and TCP/IP reference model.

   M1.04      Explain the OSI security architecture in detail.      2
Understanding

              Explain the needs for Network security and the
   M1.05                                                        2
              fundamental concepts of Network Security.
Understanding
              Explain the goals of Network security.
   M1.06      Describe the different types of Network           2
Understanding
              security.

   M1.07      Explain the basic model for Network Security.     2
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Wireless security and IP security.

This module is presented from the official syllabus scope for Network Security. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Wireless security and IP security.
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

in the internet and the various security concerns 2 Understanding in Data Link Layer. Explain the different ways for Securing

in the internet and the various security concerns 2 Understanding in Data Link Layer. Explain the different ways for Securing is an official topic in Module 2 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify in the internet and the various security concerns 2 Understanding in Data Link Layer. Explain the different ways for Securing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, in the internet and the various security concerns 2 understanding in data link layer. explain the different ways for securing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What in the internet and the various security concerns 2 Understanding in Data Link Layer. Explain the different ways for Securing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ in the internet and the various security concerns 2 Understanding in Data Link Layer. Explain the different ways for Securing □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

2 Understanding Ethernet LANs and Virtual LAN. Explain various wireless network threats and

2 Understanding Ethernet LANs and Virtual LAN. Explain various wireless network threats and is an official topic in Module 2 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Ethernet LANs and Virtual LAN. Explain various wireless network threats and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding ethernet lans and virtual lan. explain various wireless network threats and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding Ethernet LANs and Virtual LAN. Explain various wireless network threats and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 2 Understanding Ethernet LANs and Virtual LAN. Explain various wireless network threats and [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

2 Understanding wireless security measures.

2 Understanding wireless security measures. is an official topic in Module 2 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding wireless security measures. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding wireless security measures. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding wireless security measures. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 2 Understanding wireless security measures. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain Wireless Access Protocol (WAP). 1 Understanding Illustrate how to secure a Wireless LAN.

Explain Wireless Access Protocol (WAP). 1 Understanding Illustrate how to secure a Wireless LAN. is an official topic in Module 2 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Wireless Access Protocol (WAP). 1 Understanding Illustrate how to secure a Wireless LAN. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain wireless access protocol (wap). 1 understanding illustrate how to secure a wireless lan. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Wireless Access Protocol (WAP). 1 Understanding Illustrate how to secure a Wireless LAN. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Explain Wireless Access Protocol (WAP). 1 Understanding Illustrate how to secure a Wireless LAN. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Explain about Mobile device security and 3 Understanding security concerns in the Network Layer. Explain IPsec, its Applications, Benefits,

Explain about Mobile device security and 3 Understanding security concerns in the Network Layer. Explain IPsec, its Applications, Benefits, is an official topic in Module 2 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain about Mobile device security and 3 Understanding security concerns in the Network Layer. Explain IPsec, its Applications, Benefits, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain about mobile device security and 3 understanding security concerns in the network layer. explain ipsec, its applications, benefits, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain about Mobile device security and 3 Understanding security concerns in the Network Layer. Explain IPsec, its Applications, Benefits, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Explain about Mobile device security and 3 Understanding security concerns in the Network Layer. Explain IPsec, its Applications, Benefits, □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Services and different IPsec Communication 2 Understanding Modes.

Services and different IPsec Communication 2 Understanding Modes. is an official topic in Module 2 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Services and different IPsec Communication 2 Understanding Modes. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, services and different ipsec communication 2 understanding modes. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Services and different IPsec Communication 2 Understanding Modes. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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but different term.

Malayalam support: Module 2-□□ Services and different IPsec Communication 2 Understanding Modes. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Explain IPsec Protocols and IP security policy. 2 Understanding Data Link Layer security: Functionalities of Data Link layer in the network, Security Concerns in Data Link Layer, Securing Ethernet LANs, Securing Virtual LAN, Wireless network threats, wireless security measures, WAP, Securing Wireless LAN, Mobile device security . Network Layer security: Security in Network Layer, Overview of IPsec, Applications, Benefits, Services, IPsec Communication Modes, IPsec Protocols, IP security policy. Describe the functioning of most common security protocols employed at

Explain IPsec Protocols and IP security policy. 2 Understanding Data Link Layer security: Functionalities of Data Link layer in the network, Security Concerns in Data Link Layer, Securing Ethernet LANs, Securing Virtual LAN, Wireless network threats, wireless security measures, WAP, Securing Wireless LAN, Mobile device security . Network Layer security: Security in Network Layer, Overview of IPsec, Applications, Benefits, Services, IPsec Communication Modes, IPsec Protocols, IP security policy. Describe the functioning of most common security protocols employed at is an official topic in Module 2 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

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Study points

- Define or identify Explain IPsec Protocols and IP security policy. 2 Understanding Data Link Layer security: Functionalities of Data Link layer in the network, Security Concerns in Data Link Layer, Securing Ethernet LANs, Securing Virtual LAN, Wireless network threats, wireless security measures, WAP, Securing Wireless LAN, Mobile device security . Network Layer security: Security in Network Layer, Overview of IPsec, Applications, Benefits, Services, IPsec Communication Modes, IPsec Protocols, IP security policy. Describe the functioning of most common security protocols employed at using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain ipsec protocols and ip security policy. 2 understanding data link layer security: functionalities of data link layer in the network, security concerns in data link layer, securing ethernet lans, securing virtual lan, wireless network threats, wireless security measures, wap, securing wireless lan, mobile device security . network layer security: security in network layer, overview of ipsec, applications, benefits, services, ipsec communication modes, ipsec protocols, ip security policy. describe the functioning of most common security protocols employed at should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain IPsec Protocols and IP security policy. 2 Understanding Data Link Layer security: Functionalities of Data Link layer in the network, Security Concerns in Data Link Layer, Securing Ethernet LANs, Securing Virtual LAN, Wireless network threats, wireless security measures, WAP, Securing Wireless LAN, Mobile device security . Network Layer security: Security in Network Layer, Overview of IPsec, Applications, Benefits, Services, IPsec Communication Modes, IPsec Protocols, IP security policy. Describe the functioning of most common security protocols employed at means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Explain IPsec Protocols and IP security policy. 2 Understanding Data Link Layer security: Functionalities of Data Link layer in the network, Security Concerns in Data Link Layer, Securing Ethernet LANs, Securing Virtual LAN, Wireless network threats, wireless security measures, WAP, Securing Wireless LAN, Mobile device security . Network Layer security: Security in Network Layer, Overview of IPsec, Applications, Benefits, Services, IPsec Communication Modes, IPsec Protocols, IP security policy. Describe the functioning of most common security protocols employed at □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Wireless security and IP security.

            Explain the functionalities of Data Link Layer
  M2.01     in the internet and the various security concerns   2
Understanding
            in Data Link Layer.
            Explain the different ways for Securing
  M2.02                                                         2
Understanding
            Ethernet LANs and Virtual LAN.
            Explain various wireless network threats and
  M2.03                                                         2
Understanding
            wireless security measures.

  M2.04     Explain Wireless Access Protocol (WAP).             1
Understanding

            Illustrate how to secure a Wireless LAN.
  M2.05     Explain about Mobile device security and           3
Understanding
            security concerns in the Network Layer.
            Explain IPsec, its Applications, Benefits,
  M2.06     Services and different IPsec Communication         2
Understanding
            Modes.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Transport Layer and Application layer.

This module is presented from the official syllabus scope for Network Security. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Transport Layer and Application layer.
- Official duration: As specified
- Official topic entries captured: 9
- The full source excerpt is preserved below for coverage checking.

Explain the Need for Transport Layer Security. 1 Understanding Illustrate the role of Secure Socket Layer (SSL)

Explain the Need for Transport Layer Security. 1 Understanding Illustrate the role of Secure Socket Layer (SSL) is an official topic in Module 3 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the Need for Transport Layer Security. 1 Understanding Illustrate the role of Secure Socket Layer (SSL) using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the need for transport layer security. 1 understanding illustrate the role of secure socket layer (ssl) should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the Need for Transport Layer Security. 1 Understanding Illustrate the role of Secure Socket Layer (SSL) means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Explain the Need for Transport Layer Security. 1 Understanding Illustrate the role of Secure Socket Layer (SSL) □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

and Transport Layer Security(TLS). Explain 2 Understanding SSL and TLS architecture and protocol stack. Differentiate SSL and TLS.

and Transport Layer Security(TLS). Explain 2 Understanding SSL and TLS architecture and protocol stack. Differentiate SSL and TLS. is an official topic in Module 3 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify and Transport Layer Security(TLS). Explain 2 Understanding SSL and TLS architecture and protocol stack. Differentiate SSL and TLS. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, and transport layer security(tls). explain 2 understanding ssl and tls architecture and protocol stack. differentiate ssl and tls. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What and Transport Layer Security(TLS). Explain 2 Understanding SSL and TLS architecture and protocol stack. Differentiate SSL and TLS. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ and Transport Layer Security(TLS). Explain 2 Understanding SSL and TLS architecture and protocol stack. Differentiate SSL and TLS. □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

2 Understanding Discuss the various SSL/TLS attacks . Explain HTTPS-Connection initiation,

2 Understanding Discuss the various SSL/TLS attacks . Explain HTTPS-Connection initiation, is an official topic in Module 3 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

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Study points

- Define or identify 2 Understanding Discuss the various SSL/TLS attacks . Explain HTTPS-Connection initiation, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding discuss the various ssl/tls attacks . explain https-connection initiation, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding Discuss the various SSL/TLS attacks . Explain HTTPS-Connection initiation, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Malayalam support: Module 3-[illegible] 2 Understanding Discuss the various SSL/TLS attacks . Explain HTTPS-Connection initiation, [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Connection closure and Secure Shell Protocol 2 Understanding (SSH). Explain the security issues in the application

Connection closure and Secure Shell Protocol 2 Understanding (SSH). Explain the security issues in the application is an official topic in Module 3 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Connection closure and Secure Shell Protocol 2 Understanding (SSH). Explain the security issues in the application using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, connection closure and secure shell protocol 2 understanding (ssh). explain the security issues in the application should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Connection closure and Secure Shell Protocol 2 Understanding (SSH). Explain the security issues in the application means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Malayalam support: Module 3-[illegible] Connection closure and Secure Shell Protocol 2 Understanding (SSH). Explain the security issues in the application [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

1 Understanding layer, Explain the different threats that may affect

1 Understanding layer, Explain the different threats that may affect is an official topic in Module 3 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

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Study points

- Define or identify 1 Understanding layer, Explain the different threats that may affect using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding layer, explain the different threats that may affect should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding layer, Explain the different threats that may affect means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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but different term.

Malayalam support: Module 3-□□ 1 Understanding layer, Explain the different threats that may affect □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Email and the measures to ensure E mail 2 Understanding security. Explain the factors to be considered for creating

Email and the measures to ensure E mail 2 Understanding security. Explain the factors to be considered for creating is an official topic in Module 3 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Email and the measures to ensure E mail 2 Understanding security. Explain the factors to be considered for creating using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, email and the measures to ensure e mail 2 understanding security. explain the factors to be considered for creating should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Email and the measures to ensure E mail 2 Understanding security. Explain the factors to be considered for creating means in the official course context
Study lens	Principle → components or stages → result → application
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2 Understanding an e-mail policy and the steps involved.

2 Understanding an e-mail policy and the steps involved. is an official topic in Module 3 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding an e-mail policy and the steps involved. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding an e-mail policy and the steps involved. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding an e-mail policy and the steps involved. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 3-□□ 2 Understanding an e-mail policy and the steps involved. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Explain PGP and S/MIME in detail.

Explain PGP and S/MIME in detail. is an official topic in Module 3 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain PGP and S/MIME in detail. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain pgp and s/mime in detail. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain PGP and S/MIME in detail. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Explain PGP and S/MIME in detail. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Illustrate how to ensure DNS security. 1 Understanding Transport Layer security: Need for Transport Layer Security, Secure Socket Layer (SSL), TLS architecture, TLS Protocol, SSL/TLS attacks, HTTPS-Connection initiation, Connection closure, Secure Shell Protocol (SSH) . Application Layer security : E-mail Security – Email threats, Creating e-mail policy, PGP, S / MIME, DNS Security. Understand the importance of Network Access Control and the role of

Illustrate how to ensure DNS security. 1 Understanding Transport Layer security: Need for Transport Layer Security, Secure Socket Layer (SSL), TLS architecture, TLS Protocol, SSL/TLS attacks, HTTPS-Connection initiation, Connection closure, Secure Shell Protocol (SSH) . Application Layer security : E-mail Security – Email threats, Creating e-mail policy, PGP, S / MIME, DNS Security. Understand the importance of Network Access Control and the role of is an official topic in Module 3 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate how to ensure DNS security. 1 Understanding Transport Layer security: Need for Transport Layer Security, Secure Socket Layer (SSL), TLS architecture, TLS Protocol, SSL/TLS attacks, HTTPS-Connection initiation, Connection closure, Secure Shell Protocol (SSH) . Application Layer security : E-mail Security – Email threats, Creating e-mail policy, PGP, S / MIME, DNS Security. Understand the importance of Network Access Control and the role of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate how to ensure dns security. 1 understanding transport layer security: need for transport layer security, secure socket layer (ssl), tls architecture, tls protocol, ssl/tls attacks, https-connection initiation, connection closure, secure shell protocol (ssh) . application layer security : e-mail security – email threats, creating e-mail policy, pgp, s / mime, dns security. understand the importance of network access control and the role of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate how to ensure DNS security. 1 Understanding Transport Layer security: Need for Transport Layer Security, Secure Socket Layer (SSL), TLS architecture, TLS Protocol, SSL/TLS attacks, HTTPS-Connection initiation, Connection closure, Secure Shell Protocol (SSH) . Application Layer security : E-mail Security – Email threats, Creating e-mail policy, PGP, S / MIME, DNS Security. Understand the importance of Network Access Control and the role of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Illustrate how to ensure DNS security. 1 Understanding Transport Layer security: Need for Transport Layer Security, Secure Socket Layer (SSL), TLS architecture, TLS Protocol, SSL/TLS attacks, HTTPS-Connection initiation, Connection closure, Secure Shell Protocol (SSH) . Application Layer security : E-mail Security – Email threats, Creating e-mail policy, PGP, S / MIME, DNS Security. Understand the importance of Network Access Control and the role of □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Transport Layer and Application layer.

  M3.01     Explain the Need for Transport Layer Security.      1
Understanding
            Illustrate the role of Secure Socket Layer (SSL)
  M3.02     and Transport Layer Security(TLS). Explain          2
Understanding
            SSL and TLS architecture and protocol stack.
            Differentiate SSL and TLS.
  M3.03                                                         2
Understanding
            Discuss the various SSL/TLS attacks .
            Explain     HTTPS-Connection       initiation,
  M3.04     Connection closure and Secure Shell Protocol         2
Understanding
            (SSH).
            Explain the security issues in the application
  M3.05                                                          1
Understanding
            layer,
            Explain the different threats that may affect
  M3.06     Email and the measures to ensure E mail              2
Understanding
            security.
            Explain the factors to be considered for creating
  M3.07                                                          2
Understanding
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Firewalls in Network security.

This module is presented from the official syllabus scope for Network Security. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Firewalls in Network security.
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

of Network Access Control system and 3 Understanding Network access enforcement methods. Illustrate how Extensible Authentication Understanding

of Network Access Control system and 3 Understanding Network access enforcement methods. Illustrate how Extensible Authentication Understanding is an official topic in Module 4 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify of Network Access Control system and 3 Understanding Network access enforcement methods. Illustrate how Extensible Authentication Understanding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, of network access control system and 3 understanding network access enforcement methods. illustrate how extensible authentication understanding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What of Network Access Control system and 3 Understanding Network access enforcement methods. Illustrate how Extensible Authentication Understanding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ of Network Access Control system and 3 Understanding Network access enforcement methods. Illustrate how Extensible Authentication Understanding □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

1 Protocol work. Explain IEEE 802.X Port based Network

1 Protocol work. Explain IEEE 802.X Port based Network is an official topic in Module 4 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Protocol work. Explain IEEE 802.X Port based Network using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 protocol work. explain ieee 802.x port based network should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Protocol work. Explain IEEE 802.X Port based Network means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 1 Protocol work. Explain IEEE 802.X Port based Network □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Access Control and how Access Control Lists 3 Understanding for securing networks can be used. Explain Firewalls, the different types of

Access Control and how Access Control Lists 3 Understanding for securing networks can be used. Explain Firewalls, the different types of is an official topic in Module 4 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Access Control and how Access Control Lists 3 Understanding for securing networks can be used. Explain Firewalls, the different types of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, access control and how access control lists 3 understanding for securing networks can be used. explain firewalls, the different types of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Access Control and how Access Control Lists 3 Understanding for securing networks can be used. Explain Firewalls, the different types of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Access Control and how Access Control Lists 3 Understanding for securing networks can be used. Explain Firewalls, the different types of [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Firewall and Stateless and Stateful Packet 3 Understanding Filtering Firewall. Distinguish between Application Gateways and

Firewall and Stateless and Stateful Packet 3 Understanding Filtering Firewall. Distinguish between Application Gateways and is an official topic in Module 4 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Firewall and Stateless and Stateful Packet 3 Understanding Filtering Firewall. Distinguish between Application Gateways and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, firewall and stateless and stateful packet 3 understanding filtering firewall. distinguish between application gateways and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Firewall and Stateless and Stateful Packet 3 Understanding Filtering Firewall. Distinguish between Application Gateways and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Firewall and Stateless and Stateful Packet 3 Understanding Filtering Firewall. Distinguish between Application Gateways and □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

1 Understanding Circuit-Level Gateways.

1 Understanding Circuit-Level Gateways. is an official topic in Module 4 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding Circuit-Level Gateways. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding circuit-level gateways. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding Circuit-Level Gateways. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] 1 Understanding Circuit-Level Gateways. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Illustrate the Firewall Deployment with DMZ. 1 Understanding Explain the security issues in internet banking

Illustrate the Firewall Deployment with DMZ. 1 Understanding Explain the security issues in internet banking is an official topic in Module 4 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the Firewall Deployment with DMZ. 1 Understanding Explain the security issues in internet banking using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the firewall deployment with dmz. 1 understanding explain the security issues in internet banking should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the Firewall Deployment with DMZ. 1 Understanding Explain the security issues in internet banking means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Illustrate the Firewall Deployment with DMZ. 1 Understanding Explain the security issues in internet banking □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

2 Understanding system. Network Access Control: Elements of a Network Access Control system , Network access enforcement methods, Extensible Authentication Protocol, IEEE 802.X Port based Network Access Control, Access Control Lists. Firewalls: Types of Firewall, Stateless & Stateful Packet Filtering Firewall , Application Gateways , Circuit-Level Gateway, Firewall Deployment with DMZ. Security in internet banking system. Texts / References: T/R Book Title/Author Network Security Essentials: Applications and Standards, 3/e, William T1 Stallings, Pearson, ISBN: 9788131716649. Network Security Bible, 2nd edition, Eric Cole, Wiley Publisher, ISBN: T2 978812652331. Network Security and Management,2nd edition, Brijendra Sing, PHI, ISBN: R1 9788120339101 . Online Resources: Sl.No Website Link 1 https://www.tutorialspoint.com/network_security/network_security_tutorial.pdf 2 https://training.apnic.net/wp-content/uploads/sites/2/2016/12/TSEC01.pdf

2 Understanding system. Network Access Control: Elements of a Network Access Control system , Network access enforcement methods, Extensible Authentication Protocol, IEEE 802.X Port based Network Access Control, Access Control Lists. Firewalls: Types of Firewall, Stateless & Stateful Packet Filtering Firewall , Application Gateways , Circuit-Level Gateway, Firewall Deployment with DMZ. Security in internet banking system. Texts / References: T/R Book Title/Author Network Security Essentials: Applications and Standards, 3/e, William T1 Stallings, Pearson, ISBN: 9788131716649. Network Security Bible, 2nd edition, Eric Cole, Wiley Publisher, ISBN: T2 978812652331. Network Security and Management,2nd edition, Brijendra Sing, PHI, ISBN: R1 9788120339101 . Online Resources: Sl.No Website Link 1 https://www.tutorialspoint.com/network_security/network_security_tutorial.pdf 2 https://training.apnic.net/wp-content/uploads/sites/2/2016/12/TSEC01.pdf is an official topic in Module 4 of Network Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding system. Network Access Control: Elements of a Network Access Control system , Network access enforcement methods, Extensible

Authentication Protocol, IEEE 802.X Port based Network Access Control, Access Control Lists. Firewalls: Types of Firewall, Stateless & Stateful Packet Filtering Firewall , Application Gateways , Circuit-Level Gateway, Firewall Deployment with DMZ. Security in internet banking system. Texts / References: T/R Book Title/Author Network Security Essentials: Applications and Standards, 3/e, William T1 Stallings, Pearson, ISBN: 9788131716649. Network Security Bible, 2nd edition, Eric Cole, Wiley Publisher, ISBN: T2 978812652331. Network Security and Management,2nd edition, Brijendra Sing, PHI, ISBN: R1 9788120339101 . Online Resources: Sl.No Website Link 1 https://www.tutorialspoint.com/network_security/network_security_tutorial.pdf 2 https://training.apnic.net/wp-content/uploads/sites/2/2016/12/TSEC01.pdf using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding system. network access control: elements of a network access control system , network access enforcement methods, extensible authentication protocol, ieee 802.x port based network access control, access control lists. firewalls: types of firewall, stateless & stateful packet filtering firewall , application gateways , circuit-level gateway, firewall deployment with dmz. security in internet banking system. texts / references: t/r book title/author network security essentials: applications and standards, 3/e, william t1 stallings, pearson, isbn: 9788131716649. network security bible, 2nd edition, eric cole, wiley publisher, isbn: t2 978812652331. network security and management,2nd edition, brijendra sing, phi, isbn: r1 9788120339101 . online resources: sl.no website link 1 https://www.tutorialspoint.com/network_security/network_security_tutorial.pdf 2 https://training.apnic.net/wp-content/uploads/sites/2/2016/12/tsec01.pdf should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding system. Network Access Control: Elements of a Network Access Control system , Network access enforcement methods, Extensible Authentication Protocol, IEEE 802.X Port based Network Access Control, Access Control Lists. Firewalls: Types of Firewall, Stateless & Stateful Packet Filtering Firewall , Application Gateways , Circuit-Level Gateway, Firewall Deployment with DMZ. Security in internet banking system. Texts / References: T/R Book Title/Author Network Security Essentials: Applications and Standards, 3/e, William T1 Stallings, Pearson, ISBN: 9788131716649. Network Security Bible, 2nd edition, Eric Cole, Wiley Publisher, ISBN: T2 978812652331. Network Security and Management,2nd edition, Brijendra Sing, PHI, ISBN: R1 9788120339101 . Online Resources: Sl.No Website Link 1 https://www.tutorialspoint.com/network_security/network_security_tutorial.pdf 2

Aspect	What to prepare
	https://training.apnic.net/wp-content/uploads/sites/2/2016/12/TSEC01.pdf means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 2 Understanding system. Network Access Control: Elements of a Network Access Control system , Network access enforcement methods, Extensible Authentication Protocol, IEEE 802.X Port based Network Access Control, Access Control Lists. Firewalls: Types of Firewall, Stateless & Stateful Packet Filtering Firewall , Application Gateways , Circuit-Level Gateway, Firewall Deployment with DMZ. Security in internet banking system. Texts / References: T/R Book Title/Author Network Security Essentials: Applications and Standards, 3/e, William T1 Stallings, Pearson, ISBN: 9788131716649. Network Security Bible, 2nd edition, Eric Cole, Wiley Publisher, ISBN: T2 978812652331. Network Security and Management,2nd edition, Brijendra Sing, PHI, ISBN: R1 9788120339101 . Online Resources: Sl.No Website Link 1 https://www.tutorialspoint.com/network_security/network_security_tutorial.pdf 2 https://training.apnic.net/wp-content/uploads/sites/2/2016/12/TSEC01.pdf □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Firewalls in Network security.

             Explain Network Access Control, the elements
  M4.01      of Network Access Control system and                3
Understanding
             Network access enforcement methods.
             Illustrate how     Extensible   Authentication
Understanding
  M4.02                                                          1
             Protocol work.
             Explain IEEE 802.X Port based Network
  M4.03      Access Control and how Access Control Lists         3
Understanding
             for securing networks can be used.
             Explain Firewalls, the different types of
  M4.04      Firewall and Stateless and Stateful Packet          3
Understanding
             Filtering Firewall.
             Distinguish between Application Gateways and
  M4.05                                                          1
Understanding
             Circuit-Level Gateways.

  M4.06      Illustrate the Firewall Deployment with DMZ.        1
Understanding
             Explain the security issues in internet banking
  M4.07                                                          2
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 2 Understanding on the topology. Explain computer networks classification based. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding on the topology. Explain computer networks classification based*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding on the scale and the distance they cover. Explain Network Reference model, ISO/OSI. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding on the scale and the distance they cover. Explain Network Reference model, ISO/OSI*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding reference model and TCP/IP reference model.. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding reference model and TCP/IP reference model.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain the OSI security architecture in detail. 2 Understanding Explain the needs for Network security and the. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the OSI security architecture in detail. 2 Understanding Explain the needs for Network security and the*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps,

□□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Module 2

Define or explain in the internet and the various security concerns 2 Understanding in Data Link Layer. Explain the different ways for Securing. (3 marks)

Answer: Begin with the standard definition or identity of *in the internet and the various security concerns 2 Understanding in Data Link Layer. Explain the different ways for Securing*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 2 Understanding Ethernet LANs and Virtual LAN. Explain various wireless network threats and. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding Ethernet LANs and Virtual LAN. Explain various wireless network threats and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 2 Understanding wireless security measures.. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding wireless security measures.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain Wireless Access Protocol (WAP). 1 Understanding Illustrate how to secure a Wireless LAN.. (3 marks)

Answer: Begin with the standard definition or identity of *Explain Wireless Access Protocol (WAP). 1 Understanding Illustrate how to secure a Wireless LAN.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain Explain the Need for Transport Layer Security. 1 Understanding Illustrate the role of Secure Socket Layer (SSL). (3 marks)

Answer: Begin with the standard definition or identity of *Explain the Need for Transport Layer Security. 1 Understanding Illustrate the role of Secure Socket Layer (SSL)*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain and Transport Layer Security(TLS). Explain 2 Understanding SSL and TLS architecture and protocol stack. Differentiate SSL and TLS.. (3 marks)

Answer: Begin with the standard definition or identity of *and Transport Layer Security(TLS). Explain 2 Understanding SSL and TLS architecture and protocol stack. Differentiate SSL and TLS.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 2 Understanding Discuss the various SSL/TLS attacks . Explain HTTPS-Connection initiation,. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding Discuss the various SSL/TLS attacks . Explain HTTPS-Connection initiation,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Connection closure and Secure Shell Protocol 2 Understanding (SSH). Explain the security issues in the application. (3 marks)

Answer: Begin with the standard definition or identity of *Connection closure and Secure Shell Protocol 2 Understanding (SSH). Explain the security issues in the application*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Module 4

Define or explain of Network Access Control system and 3 Understanding Network access enforcement methods. Illustrate how Extensible Authentication Understanding. (3 marks)

Answer: Begin with the standard definition or identity of *of Network Access Control system and 3 Understanding Network access enforcement methods. Illustrate how*

Extensible Authentication Understanding. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 1 Protocol work. Explain IEEE 802.X Port based Network. (3 marks)

Answer: Begin with the standard definition or identity of *1 Protocol work. Explain IEEE 802.X Port based Network*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Access Control and how Access Control Lists 3 Understanding for securing networks can be used. Explain Firewalls, the different types of. (3 marks)

Answer: Begin with the standard definition or identity of *Access Control and how Access Control Lists 3 Understanding for securing networks can be used. Explain Firewalls, the different types of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Firewall and Stateless and Stateful Packet 3 Understanding Filtering Firewall. Distinguish between Application Gateways and. (3 marks)

Answer: Begin with the standard definition or identity of *Firewall and Stateless and Stateful Packet 3 Understanding Filtering Firewall. Distinguish between Application Gateways and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **2 Understanding on the topology. Explain computer networks classification based**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **in the internet and the various security concerns 2 Understanding in Data Link Layer. Explain the different ways for Securing**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Explain the Need for Transport Layer Security. 1 Understanding Illustrate the role of Secure Socket Layer (SSL)**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **of Network Access Control system and 3 Understanding Network access enforcement methods. Illustrate how Extensible Authentication Understanding**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

- T/R Book Title/Author
- Network Security Essentials: Applications and Standards, 3/e, William
- Stallings, Pearson, ISBN: 9788131716649.
- Network Security Bible, 2nd edition, Eric Cole, Wiley Publisher, ISBN:
- Network Security and Management,2nd edition, Brijendra Sing, PHI, ISBN:
- Online Resources:
- Sl.No Website Link
- https://www.tutorialspoint.com/network_security/network_security_tutorial.pdf
- https://training.apnic.net/wp-content/uploads/sites/2/2016/12/TSEC01.pdf

Web resources listed in the official PDF

- Sl.No Website Link
- https://www.tutorialspoint.com/network_security/network_security_tutorial.pdf
- https://training.apnic.net/wp-content/uploads/sites/2/2016/12/TSEC01.pdf

IN-PAGE SEARCH

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Prepared from the supplied official SITTTR syllabus PDF for Course 5282. Verify institutional updates against the current official curriculum.